Summer 25

Q2

#	Data Length	MF	DF	Offset
…	…	…	…	…
10	…	1	1	1854
…	…	…	…	…
15	69	0	1	2884

Given, the header size used in this is 38 bytes. From the above table,

[CO3] **I.** **Identify** the packet size of the 10th fragment

[CO3] **II.** **Identify** the data size of the original datagram

[CO2] **III.** For the offset field, we convert the 16 bit number to 13 bits. Despite shortening it by three bits, how are we preventing any data loss?

4 + 3 + 2

I. Header size = 38 B

Packet size = Data + Header

At 10th fragment offset = 1854 and MF = 1

So, before it 9 fragment have done.

$$\text{offset} = \frac{\text{total data proceed}}{8} \times 9$$

$$\text{Data size} = \frac{1854 \times 8}{9} = 1648 \text{ B}$$

$$\text{Packet size} = 1648 + 38 = 1686 \text{ B}$$

II. Data size = 1648

Total 14 full fragment and last additional 69 B

$$\text{Total data size} = (1648 \times 14) + 69$$
$$= 23141 \text{ B}$$

III. Because last 3 bit position is considered as 000. So dividing by 8 or right shift 3 bit prevent any data loss.

Q2 4 + 3 + 2

#	**Total Length**	**MF**	**DF**	**Offset**
…	…	…	…	…
10	1544	1	1	1692
…	…	…	…	…
15	69	0	1	2632

From the above table,

[CO3] **I. Identify** the header size of the fragment

[CO3] **II. Identify** the packet size of the original datagram

[CO2] **III.** Which two fields are required to orderly reassemble the fragmented packets?

I. At 10th fragment :

Total Length = 1544 , MF = 1

offset = 1692

$$\text{offset} = \frac{\text{Total data}}{8} \times 9 \quad [\text{9 fragment done}]$$

$$\text{Total data} = \frac{1692 \times 8}{9} = 1504 \text{ B}$$

Total Length = Data + Header

Header = 1544 − 1504 = 40 B

II. Data size = 1504 B

Total Packet size = (1504 × 14) + 69 = 21125 B

iii. Identification and fragment offset.

Q3	A packet of size **2584 bytes** including **32 bytes** of header was fragmented for transmitting in a link with MTU=**X bytes**. The **9th** packet has a size of **272 bytes.** It also has its MF bit set.	2 + 3 + 3
[CO3]	**I.** **Calculate** the value of X.	
[CO3]	**II.** **Calculate** the fragment offset for the 7th packet.	
[CO3]	**III.** **Find out** the total number of fragmented packets.	

I. As the 9th packet size = 272 bytes and MF = 1 means the packet must be in full size and more fragment to come.

So, MTU = 272 bytes

II. Total size = 272, Header = 32, Data = 272 − 32 = 240

Before 7th packet there are 6 packets.

So, total preceding byte = 240 × 6 = 1440 byte

$$\text{offset} = \frac{1440}{8} = 180$$

The fragment offset of the 7th packet = 180

III. Original Packet size = 2584

Header = 32

Total Data = 2584 − 32 = 2552 bytes

240 byte data can fit per full segment.

$$240 \,)\, 2552 \,(\, 10$$
$$\underline{2400}$$
$$152$$

So, 10 full packet and 1 for extra 152B of data.

Total number of fragmented packet = 10+1

= 11

Q3		A packet of size 4584 bytes including 42 bytes of header was fragmented for transmitting in a link with MTU=X bytes. The 9th packet has a size of 362 bytes. It also has its MF bit set.	2 + 3 + 3
[CO3]	I.	Calculate the value of X.	
[CO3]	II.	Calculate the fragment offset for the 5th packet.	
[CO3]	III.	Find out the total number of fragmented packets.	

1. As the 9th packet size = 362 bytes and MF = 1 means the packet must be in full size and more fragment to come.

So, MTU = 362 bytes

ii. Total size = 362, Header = 42, Data = 362 - 42 = 320

Before 5th packet there are 4 packets.

So, total preceding byte = 320 x 4 = 1280 bytes

$$\text{offset} = \frac{1280}{8} = 160$$

The fragment offset of the 5th packet = 160

iii. Original Packet size = 4584

Header = 42

Total Data = 4584 - 42 = 4542 bytes

320 byte data can fit per full segment.

$$\begin{array}{r|l} 320 & 4542 \;|\; 14 \\ & \underline{4480} \\ & \;\;\;62 \end{array}$$

So, 14 full packet and 1 for extra 62B of data.

Total number of fragmented packet = 14+1

= 15

Fall 24

	Q3	An IPv4 packet is received at the end of the link with header parameters set as: *Version = 4, IHL = 5, TOS = 0, Total Length = 5086, Identification = 5656, DF = 0, MF = 0, Fragmentation Offset = 0, TTL = 45, Protocol = 17* The router that received the packet identified that **1244 Bytes** is the maximum packet size that can be successfully sent via the link. [IPv4 header is **20 bytes** in length]	3 + 3 + 3 + 3 + 2
[CO3]	**I.**	**Identify** the number of fragments that will be created.	
[CO3]	**II.**	**Calculate** the fragment size of the last packet.	
[CO3]	**III.**	**Identify** the fragment offset of the **5th** fragment if the initial byte number was set to 0.	
[CO2]	**IV.**	**Explain** the significance of the Identification field.	
[CO2]	**V.**	**Find out** what the router would do if the DF was 1.	

I. Total length = 5086 B

Header Size = 20 B

Data = 5086 − 20 = 5066 B

Max. packet size = 1244 B

Data per fragment = 1244 − 20 = 1224

total fragment = 1224) 5066 (4
4896
170

= 4 + 1 (extra 170)

= 5

II. Remaining data after 4th packet = 170

Fragment size of last packet = 170 + 20
↳ Header

= 190 B

III. Data send before 5th fragment = 4 × 1224 B

= 4896 B

Fragment offset = $\frac{4896}{8}$ = 612

IV. ID allows the destination to to recognize which fragments belong together to reassemble the original data correctly.

V. DF = 1 means do not fragment so router can not fragment the oversize packet so it will discard the packet and send an error message back to sender.

Q3		An IPv4 packet is received at the end of the link with header parameters set as: *Version = 4, IHL = 6, TOS = 0, Total Length = 6421, Identification = 5656, DF = 0, MF = 0, Fragmentation Offset = 0, TTL = 45, Protocol = 17* The router that received the packet identified that **1624 Bytes** is the maximum data size that can be successfully sent via the link. [IPv4 header is **24 bytes** in length]	3 + 3 + 3 + 3 + 2
[CO3]	I.	**Identify** the number of fragments that will be created.	
[CO3]	II.	**Calculate** the data size of the last packet.	
[CO3]	III.	**Identify** the fragment offset of the **last** fragment if the initial byte number was set to 0.	
[CO2]	IV.	**Explain** how the packets are re-grouped back when the fragments reach the destination.	
[CO2]	V.	**Find out** the significance of the MF flag.	

1. Total Length = 6421

Header size = 24 B

Data = 6421 - 24 = 6397 B

Max. data size per Fragment = 1624 B

total fragment = 1624 | 6397 | 3

4872

1525

= 3 + 1 (extra)

= 4

ii. Remaining data After 3rd segment is 1525.

Data size of last packet = 1525 B

iii. Last fragment starts after 4872 bytes of data.

$$\text{offset} = \frac{4872}{8} = 609$$

iv. By using Identification field and correct seq. using the fragment offset value.

v. MF tells if there are more fragment to come (MF = 1) or not (MF = 0).

Summer 24

Q3		A packet of size **7240 bytes** including **40 bytes** of header arrive at a router. The router can send at most **800 bytes** at a time through the link and thus fragment accordingly.	2 + 3 + 4 + 3
[CO3]	I.	**Calculate** the number of fragments that will be created.	
[CO3]	II.	**Calculate** the fragment size of the last fragment.	
[CO3]	III.	**Calculate** the fragment offset of the 8th fragment.	
[CO2]	IV.	**Explain** why the MF bit is always zero for the last fragment.	

I. Packet size = 7240 B

Header size = 40 B

MTU = 800 B, Data = 800 − 40 = 760 B

Total Data size = 7240 − 40 = 7200 B

Fragment = 760) 7200 (9

$$\begin{array}{r} 7200 \\ \underline{6840} \\ 360 \end{array}$$

total fragment = 9 + 1 (for 360) = 10

II. Data after 9th segment = 360 B

Fragment size = 360 + 40 [header]

= 400 B

III. $$\text{offset} = \frac{\text{total data up to 7 fragment}}{8}$$

$$= \frac{760 \times 7}{8} = 665$$

IV. MF = 0 means no more segment been done as this is the very last piece of data.

Q3	A packet with data of **8240 bytes** arrived at a router. The router can send **830 bytes** (including **30 bytes** of header) at a time through the link, so it needs to fragment accordingly.	2 + 3 + 4 + 3
[CO3]	**I. Calculate** the number of fragments that will be created.	
[CO3]	**II. Calculate** the data size of the last fragment.	
[CO3]	**III. Calculate** the fragment offset of the 7th fragment.	
[CO2]	**IV. Explain** what the router will do if the Don't Fragment (DF) bit is turned on.	

I. Data = 8240 B , Header = 30 B

MTU = 830 , Data = 830 − 30 = 800 B

fragment = 800 | 8240 | 10
8000
240

total fragment = 10 + 1 = 11

II. Data size of last fragment = 240 B

iii. offset of 7th segment = $\frac{\text{Data up to 6th}}{8}$

$= \frac{800 \times 6}{8}$

$= 600$

IV. DF 1 means router drop packet and send error message back to sender as packet exceeds MTU but not able to being fragmented.